Yutong Chen

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Education

Massachusetts Institute of Technology 2025 - 2030 (expected)

Doctor of Philosophy in Mathematics

- Research Interest: geometric representation theory.

Trinity College, University of Cambridge 2024 - 2025

Master of Advanced Study in Pure Mathematics *Classification: Distinction*

- Final Mark: 90.00%
- Master Thesis: *Schubert Varieties and Frobenius Splitting* (supervised by Prof. Rong Zhou).

Oriel College, University of Oxford 2021 - 2024

Bachelor of Arts in Mathematics *Classification: First Class*

- Final Mark: 91.00
- Ranked top 2 all three years.

Publications and Preprints

Spin Representations of Finite Coxeter Groups and Generalisations of Saxl's Conjecture

- Preprint: arXiv 2409.17540, joint with Felix Gu and Will Osborne.

Curtain Model for CAT(0) Spaces and Isometries

- Geom Dedicata 219, 69 (2025). https://doi.org/10.1007/s10711-025-01031-4.

Awards and Bursaries

Trinity Studentship in Mathematics 2024

- Full scholarship (£53,667) awarded to a mathematics master's student at Trinity College, Cambridge. This scholarship covers all tuition fees and living costs

Junior Mathematical Prize 2024

- Awarded for top 2 performance in Part B (Year 3 Exam) at Oxford

LMS Bursaries for Undergraduate Research 2024

- £3581.66 granted by the London Mathematical Society and University of Oxforrd to fund a summer research project under the supervision of Prof. Dan Ciubotaru at the Mathematical Institute, Oxford

Gibbs Prize in Mathematics 2023

- Awarded for the best performance in Part A (Year 2 Exam) at Oxford

Departmental Funding for Summer Research Program 2023

- £3,733.18 awarded by University of Oxford for the undergraduate summer research program

Services

Yau Math Camp July, 2026

Teaching Assistant *Shanghai, China*

- Served as teaching assistant for Yau mathcamp for high school students at SIMIS.
- Held problem-solving sessions and mentored students for research project.

MIT Directed Reading Program January, 2026

Mentor *Cambridge, US*

- Mentored two undergraduate students on studying étale fundamental groups.

Undergraduate Seminar 2022 - 2024

Organizer *Oxford, UK*

- Organized an undergraduate seminar with 14 active members. Met weekly.

Mathematics Tutor 2021 - 2024

- Supervised students online at various levels, including A-Level maths and high school math Olympiad *Online*

Talks

Perverse Sheaves and Perverse Filtration 2026
MIT Algebraic Geometry Learning Seminar *Cambridge, US*

Beilinson-Drinfeld Quantization: Main Theorem. 2026
MIT Representation Theory Learning Seminar *Cambridge, US*

Descent Obstructions 2026
MIT Stage Seminar *Cambridge, US*

Étale Sites 2025
MIT Stage Seminar *Cambridge, US*

Generalizations of Saxl's Conjecture 2024
Oxford Algebra Seminar *Oxford, UK*

Teaching

MIT Course 18.705: Commutative Algebra 2026
Teaching Assistant

Languages and Skills

Languages: English (advanced), Chinese (native)
IT Skills: MATLAB, LaTeX, C++
Sports: Member of the Oriel College Football Club.